

What Marvista (by Salesforce Research) is:

A human-AI news reader that guides you through the whole flow — **plan** → **read** → **reflect** — using NLP to (a) fit reading to your **time budget**, (b) surface **question-driven** passages (They use **MixQG** (a T5-based model) to create likely, answerable questions from the article), (c) ask **AI prompts** while you read, and (d) produce an **explainable human-AI summary** at the end → **extractive summarizer: BERT embeddings + K-Means** sentence clustering; select sentences near cluster centroids until the target compression is met..

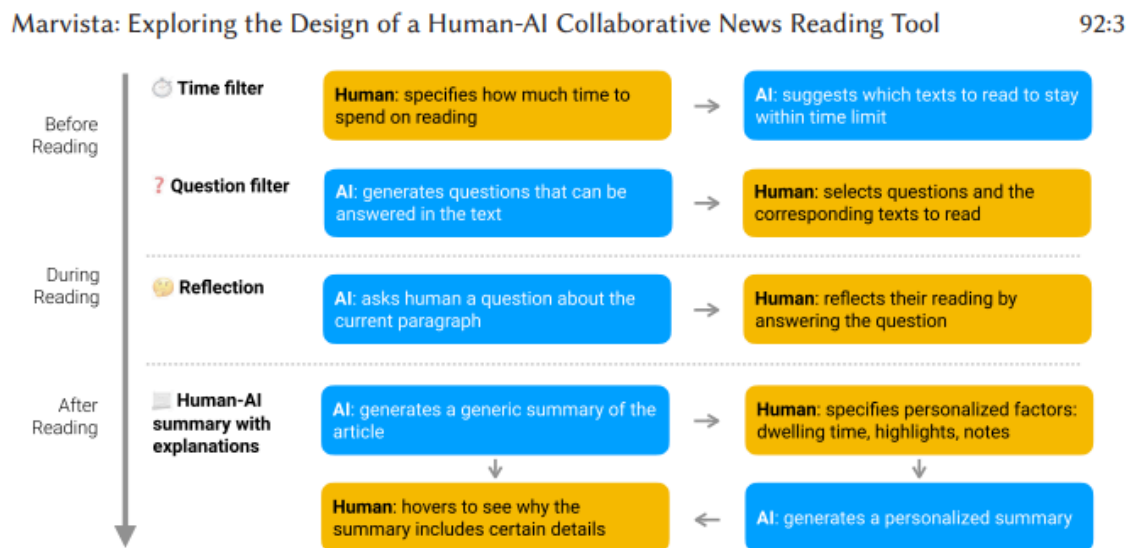


Fig. 1. We explore the design of Marvista—a human-AI collaborative tool that supports the entire process of reading an online article. Before reading: Marvista helps a user plan what to read by filtering text based on how much time one can spend (time filter) and what questions one is interested to find out (question filter); During reading: Marvista helps the user reflect on their understanding of each paragraph with an AI-generated question (reflection); After reading: Marvista generates a human-AI summary that combines AI’s processing of the text, the user’s reading behavior, and user-generated data; further, hovering over a sentence explains which paragraph it is related to and why its information is included in the summary.

CARE (Collaborative AI-Assisted Reading Environment):

it logs reading behavior and plugs into **model-agnostic NLP “assistants”** (QA, summarization, classifiers) via a broker, so you can study/support reading in real time.

Making the logs so that an NLP assistant (QA, summarizer, classifiers) can actually change how people read/comment. Logs enable before/after or A/B comparisons of behavior and outcomes.

Paper Plain CHI 23:

reading UI for medical papers that helps non-experts by:

1. showing **definitions** inline (Detect terms with **scispaCy NER** → **entity linking** to **UMLS** + **Wiktionary**.),
2. giving **section gists** (plain-language summaries),
3. **paper-aware question list**, and
4. jumping to **answering passages** with a short **answer gist**.

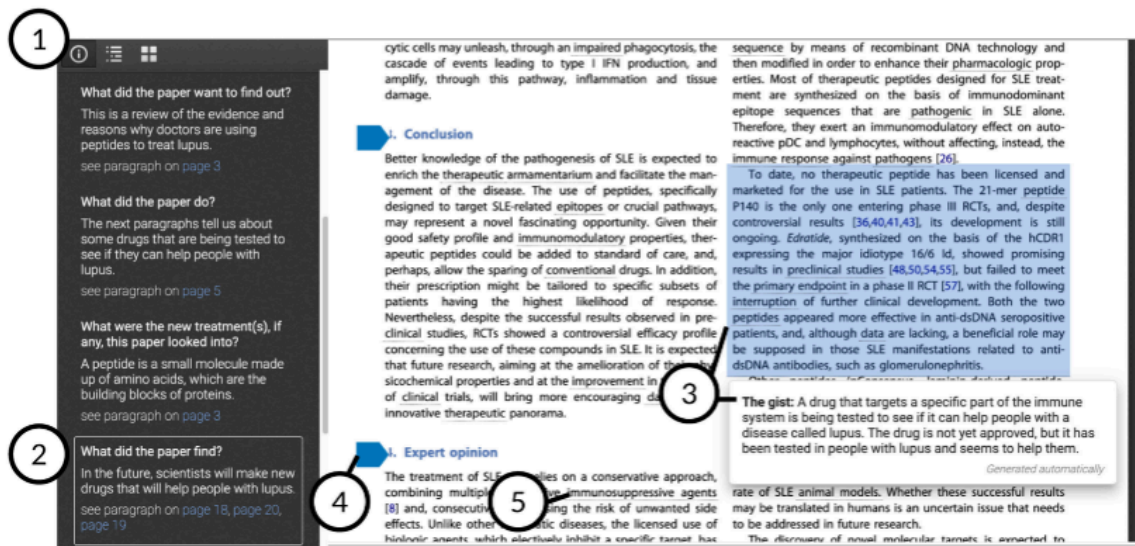


Fig. 1. Pictured is the PAPER PLAIN user interface. When a reader opens a paper in PAPER PLAIN, they see a side pane containing a reading guide (1), consisting of key questions the reader might ask of the paper, brief generated plain language answers, and pointers to passages in the paper where the reader can read more. When a reader clicks a question (2), the paper jumps to the passage that provides that answer and shows a paragraph-length plain language answer (3). Plain language summaries can be accessed for any section of the paper by clicking a label next to the section header (4). The reader can view definitions of medical terminology by clicking underlined terms (5).